

Machine-dependent nonlinear finite element responses via a crowdsourced Abaqus benchmark

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Abstract

This short note reports on a crowdsourced benchmark performed in 2026 that demonstrates different non-linear equilibrium path families arising from serial execution of identical Abaqus input files on different machines and installations. The benchmark is a cylindrical shell under uniform meridional compression, with a control case (plastic collapse) that was expected to be insensitive to the execution platform and an interesting case (elastic buckling) which was expected to be highly platform sensitive. Three dominant response families arise, aligned with CPU vendor and Abaqus release, but only one follows the expected fundamental path over the computed range. The variability when quantified using pairwise Fréchet distances appears genuine, but the crowdsourced dataset is too small and self-selected to support a causal attribution. We release the benchmark and anonymised data in the hope of starting a discussion and facilitating a larger controlled cross-platform and cross-solver experiment covering additional benchmarks at a future time.

Keywords: computational reproducibility, nonlinear finite element analysis, shell buckling, crowdsourced benchmark, Fréchet distance, cluster analysis, Abaqus

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1. The benchmark

It all started with a simple task, to execute an Abaqus input file unchanged on a different machine. The task was part of an undergraduate assignment in computational shell buckling, where students were modelling a cylindrical shell under uniform meridional compression using increasingly sophisticated finite element analyses. The first task was a materially nonlinear but geometrically linear analysis (MNA) under an elastic-perfectly plastic material model, which was to give an initial linear path followed by an indefinite plateau at the plastic collapse load. A collapse mechanism should form first through plastic hinging in the boundary layers, followed by the spread of membrane yielding through the rest of the shell (Fig. 1, left). The second task was an unperturbed geometrically nonlinear but elastic analysis (GNA) intended to trace the fundamental equilibrium path of this system. Less obviously (Section 9.5a in [1]), this response should likewise form a plateau as the bending boundary layers grow without bound and fully elastic sinusoidal deformations penetrate into the shell (Fig. 1, right). No formal imperfections were introduced. Things got interesting when students obtained neither expected response, only non-physical and furthermore machine-specific results.

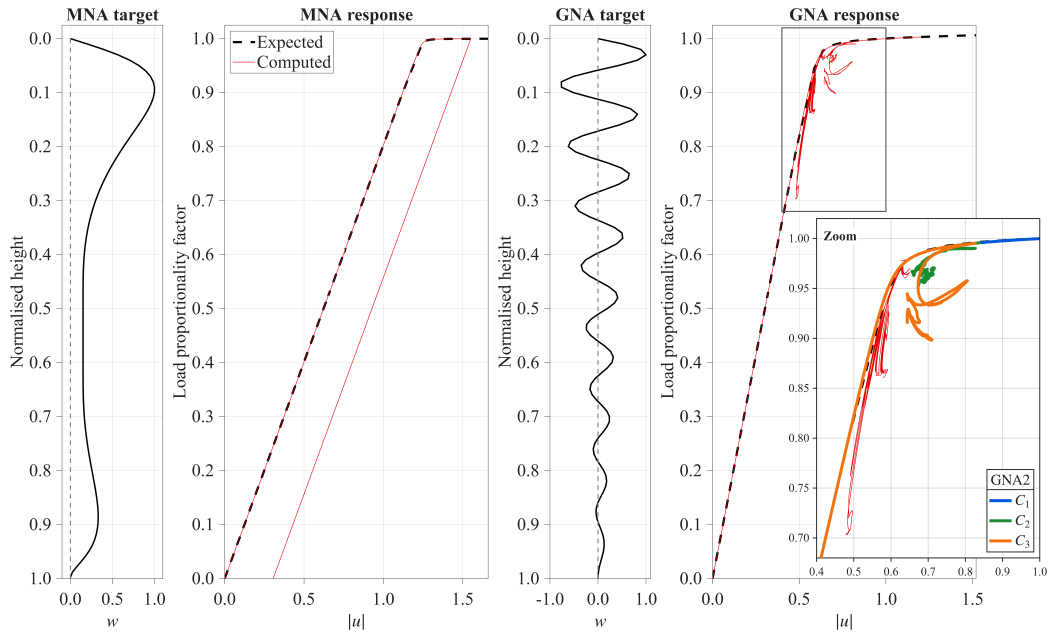


Figure 1: Target deformation profiles and corresponding response paths for the materially nonlinear analysis (MNA) and geometrically nonlinear analysis (GNA) benchmark cases. The target panels show radial deformation w over the normalised height. Response curves show the load proportionality factor against the magnitude of the loaded-edge meridional shortening $|u|$. The inset enlarges the near-peak response region and GNA2 clusters.

19 The idea of a systematic reproducibility benchmark crystallised from this
 20 experience, and a public call to crowdsource a larger dataset was made. The
 21 benchmark consisted of four small, near-identical Abaqus input files. One
 22 was an MNA ‘control’ which was not intended to be interesting, and three
 23 GNA files differing only by the starting and maximum load increment size.
 24 Participants were asked to run all files to completion, serially and without
 25 modification, then execute a Python script to extract loading histories, and
 26 document the hardware (CPU, memory) and software (Abaqus build, OS)
 27 environments in a metadata questionnaire.

28 2. Results

29 Only 33 submissions were received (despite a much larger submission
 30 target) whose distinct load histories are illustrated in Fig. 1. All MNAs
 31 followed the correct initial linear response, plateaued partially at the expected
 32 load level and then erroneously reversed the direction of loading. This reversal
 33 is a manifestation of the ‘doubling back’ pathology [2] known to occur in
 34 arc-length nonlinear solvers, but at least it was obtained consistently by all
 35 submissions. The GNAs, however, despite following the correct initial near-
 36 linear response eventually diverged into multiple macroscopically different
 37 late-stage paths, *most* of them incorrect. The fundamental GNA equilibrium
 38 becomes energetically unstable after the first bifurcation point, which for the
 39 benchmark’s boundary conditions should occur at a normalised load level
 40 approaching 0.9 (Figure 2.44 in [4] for $Z \approx 10^3$, C1-S3). Absent an explicit
 41 perturbation of the initial geometry (the standard technique for triggering
 42 bifurcations), or some other source of perturbation, the solver is expected to
 43 trace this smooth, unstable fundamental path correctly.

44 Deterministic SHA256 fingerprints identified 12, 11, 14 and 11 unique his-
 45 tories for MNA, GNA1, GNA2 and GNA3, respectively. Pairwise multivariate
 46 Fréchet distances [3] were computed between these unique histories and are
 47 shown as colour-coded matrices in Fig. 2 for the MNA and GNA2 cases only.
 48 MNA histories, intended to be an exactly reproducible control, unexpectedly
 49 contained microscopic differences. That these still produce the same (albeit
 50 erroneous) macroscopic plastic collapse response is consistent with MNAs
 51 being perturbation insensitive, for this system at least. By contrast, the GNA
 52 responses exhibit severe variability, with microscopic differences cascading to
 53 give macroscopically radically different histories (Fig. 1, right), apparently
 54 from platform-specific numerical causes. This is consistent with the known
 55 sensitivity of thin cylinders under meridional compression to imperfections
 56 (Sections 4.6 & 4.7 in [4]), and generally with the sensitivity of unstable
 57 nonlinear systems to perturbation.

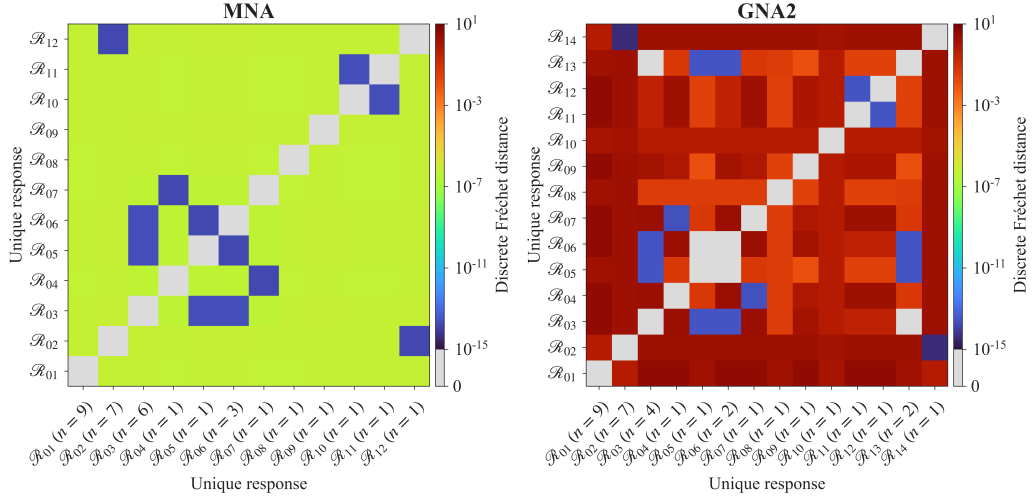


Figure 2: Pairwise discrete Fréchet distances between the unique MNA and GNA2 response histories represented by the standardised load proportionality factor, loaded-edge meridional shortening and maximum force residual, with arc length retained implicitly through point ordering. Colours show distance on a common logarithmic scale, grey denoting zero. n is the number of submissions for each unique response $\mathcal{R}_i$. At least part of the microscopic variability may still be attributable to export-formatting effects.

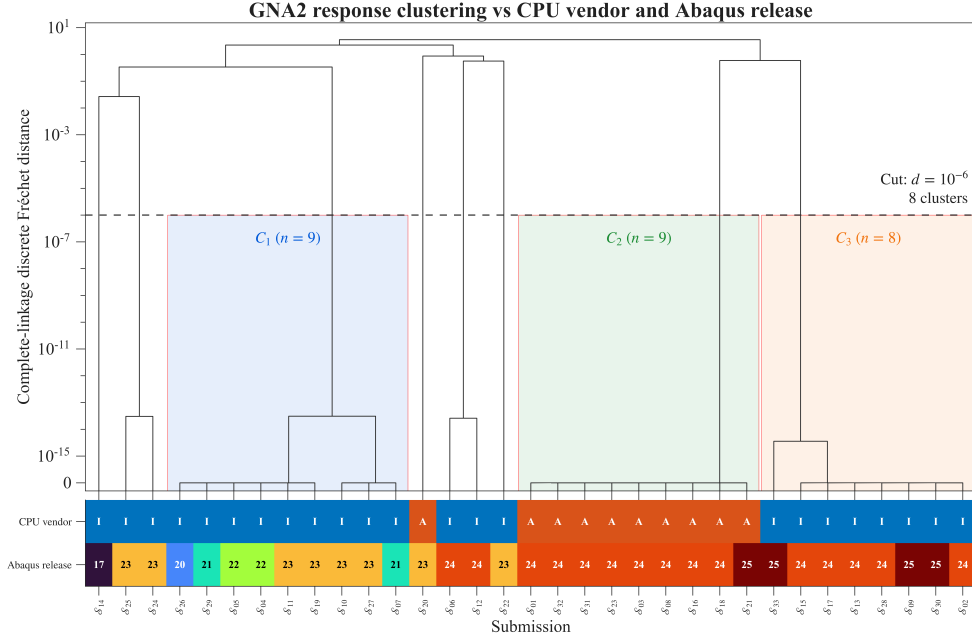


Figure 3: Dendrogram of GNA2 response histories based on pairwise multivariate Fréchet distances with complete linkage. Submissions $\mathcal{S}_i$ are labelled by CPU vendor and Abaqus release. A reasonable cut at $d = 10^{-6}$ highlights three dominant clusters, two doubletons and three singletons. Responses of clusters C_1 , C_2 and C_3 are highlighted in Fig. 1, inset.

Recurring GNA paths indicate structured, partly reproducible variability. Complete-linkage clustering favouring internally compact clusters suggests three dominant families of submissions in the dataset: Intel CPUs with older Abaqus versions 2020-23 (C_1), AMD CPUs with newer Abaqus versions 2024-25 (C_2) and Intel CPUs with newer Abaqus versions (C_3). The clusters are illustrated in Fig. 3 for GNA2 only but recur across all analyses. Of these, only C_1 for GNA2 actually follows the expected response (Fig. 1, inset).

3. Conclusions

This short note reports on a crowdsourced benchmark that illustrates how unchanged Abaqus input files produced recurring and macroscopically distinct nonlinear responses correlated with the computational platform. The authors call for a larger controlled cross-platform and cross-solver reproducibility study on more safety-critical structural systems of industrial significance to raise awareness and accurately identify correlations and causal mechanisms.

Declarations

CRedit authorship statement. Yuhan Wei: Formal analysis, Writing - original draft. Molly Balassa: Conceptualization, Formal analysis, Writing - original draft. Adam J. Sadowski: Conceptualization, Formal analysis, Writing - original draft. **Data and code availability.** Anonymised data, benchmark materials and processing code are available at https://github.com/SadowskiAJ/RRX_public.git. There is significantly more in the data than has been reported on here. **Declaration of generative AI use.** OpenAI Codex was used to assist with development of the analysis code. All resulting code was reviewed by the authors. **Competing interests.** The authors declare no competing interests. **Acknowledgements.** The authors gratefully acknowledge the participation of all crowdsourcing contributors. Dr Sadowski acknowledges EPSRC Fellowship funding under EP/Z533622/1, Mx Balassa acknowledges an EPSRC ICASE studentship.

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